UNIVERSITY OF MAKATI • COLLEGE OF COMPUTING AND INFORMATION SCIENCES

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MARKET SIZING (TAM • SAM • SOM) & PRODUCT COSTING REPORT

Application of Chapter 3 Concepts to Technopreneurship (Hulog) and Capstone Projects

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Techno Project:	**HULOG:** IoT Dynamic QR/PIN Parcel Drop Vault	**Capstone Project:**	Closed-Loop WAN Self-Healing Framework

01 REVIEW OF CHAPTER 3: TAM, SAM, AND SOM FRAMEWORK

In Technopreneurship, assessing the commercial viability of a technical innovation requires defining how large the real opportunity is and how much of it we can realistically capture. Chapter 3 outlines the three-tiered market sizing framework:

TAM • TOTAL ADDRESSABLE MARKET

100% Demand

The overall revenue opportunity available if our product achieved 100% market share with zero competitors and unrestricted distribution nationwide.

SAM • SERVICEABLE ADDRESSABLE MARKET

Target Segment

The portion of TAM that fits our specific product form factor, geographic reach, income class, and distribution channels (e.g., Mega Manila gated homes).

SOM • SERVICEABLE OBTAINABLE MARKET

Realistic Capture

The realistic market share we can capture over the first 1 to 3 years, constrained by production capacity, initial capital, and early marketing beachheads.

Engineering Rationale for Product Pricing: Market sizing cannot be separated from unit economics. Before calculating TAM/SAM/SOM value, we must determine the **Target Selling Price** through bottom-up expense accounting: *Total Unit Cost (COGS) = Direct Materials (BOM) + Direct Labor (Assembly/QA) + Manufacturing Overhead*, followed by a viable commercial gross profit markup (40%–55%).

02 APPLICATION TO TECHNO PROJECT: HULOG (SMART PARCEL DROP VAULT)

Product Summary: An IoT-enabled perimeter gate parcel vault designed for residential townhouses, dormitories, and boarding houses. Using an ESP32 microcontroller, a 12V 1000N solenoid lock, and an internal optical IR sensor, recipients generate a single-use, 6-digit PIN or dynamic QR code that invalidates immediately after the delivery rider deposits the package and closes the door.

A. ITEMIZED COST & EXPENSE BREAKDOWN (COGS PER UNIT)

SUBSYSTEM / ITEM	TECHNICAL SPECIFICATION & COMPONENT PURPOSE	UNIT COST (PHP)
ESP32 DevKit V1	Dual-Core 240MHz, 4MB Flash, Wi-Fi & BLE; runs FreeRTOS control loops	₱240.00
12V Solenoid Latch	12V 0.8A heavy-duty cabinet electromagnetic tongue lock (1000N force)	₱190.00
Power Subsystem	12V 2A DC Adapter (₱115) + LM2596 Buck Converter (₱55) + DC Jacks (₱20)	₱190.00
User Display & Keypad	0.96" I2C OLED SSD1306 (₱110) + 4x4 Weatherproof Membrane Keypad (₱65)	₱175.00
Sensors & Relay	MC-38 Magnetic Door Switch (₱55) + 1-Ch Relay (₱45) + Optical IR Sensor (₱25)	₱125.00
Audio Alert & Wiring	5V Active Buzzer (₱20) + Wire harnesses, terminal blocks & connectors (₱60)	₱80.00
	Subtotal — Core Electronics Subsystem:	**₱1,000.00**
Sheet Metal Enclosure	1.2mm Galvanized Iron (GI) steel box with anti-theft baffle (laser cut)	₱500.00
Powder Coating Finish	Electrostatic anti-rust, weather-resistant outdoor finish	₱120.00
Mechanical Hardware	Stainless continuous piano hinge, rubber door seals & security screws	₱80.00
	1. TOTAL DIRECT MATERIALS (BOM):	**₱1,700.00**
Direct Labor (Manpower)	2.0 hours technician labor @ ₱90.00/hr (Wiring: ₱67.50, Assembly: ₱67.50, QA: ₱45.00)	₱180.00
Manufacturing Overhead	Packaging box & foam (₱65) + consumables (₱25) + shop electricity/tool wear (₱30)	₱120.00
	TOTAL COST OF GOODS SOLD (COGS) PER UNIT:	**₱2,000.00**

B. TARGET PRICE DETERMINATION & PROFIT MARGINS

COMMERCIAL MODEL	TARGET PRICE	UNIT COGS	GROSS PROFIT	GROSS MARGIN
Hulog Core (Standard Turnkey Unit)	**₱3,499.00**	₱2,000.00	**₱1,499.00**	**42.8%**
Hulog Pro (w/ Li-Ion Battery Backup)	**₱4,499.00**	₱2,450.00	**₱2,049.00**	**45.5%**
Boarding House SaaS Portal (Monthly)	**₱199.00/mo**	₱25.00/mo	**₱174.00/mo**	**87.4%**

C. MARKET SIZING CALCULATIONS (HULOG)

TAM • NATIONWIDE

₱38.5 Billion

11,000,000 e-commerce households in PH × **₱3,500** retail price. Basis: ~42% of 26.4M PSA households regularly buy online.

SAM • MEGA MANILA

₱5.25 Billion

1,500,000 gated houses, townhouses, and dorms in NCR + CALABARZON (Class B/C with home Wi-Fi) × **₱3,500**.

SOM • 3-YEAR TARGET

₱35.0 Million

10,000 units (~0.65% SAM share) × ₱3,500 hardware. Plus ~1,200 dorm subscriptions = **+ ₱2.86M/yr SaaS**.

03 APPLICATION TO CAPSTONE PROJECT: MULTI-BRANCH HYBRID WAN SELF-HEALING FRAMEWORK

Product Summary: An enterprise network infrastructure and telemetry software suite designed for distributed public agencies and enterprises (e.g., regional offices, BPOs). It integrates edge Prometheus probe exporters, a centralized decision controller, and automated MAPE-K closed-loop routing failover to autonomously mitigate brownouts and ISP degradation without requiring a full SD-WAN rip-and-replace.

A. PRODUCTION & DEPLOYMENT COST BREAKDOWN (PER BRANCH APPLIANCE & CONTROLLER PACKAGE)

Unlike a mass-market hardware consumer gadget, enterprise network software is sold as a **deployment package (On-Prem Edge Probe Appliance + Central Controller License + 1-Year SLA Support)** for branch deployments:

EXPENSE CATEGORY	SCOPE, TOOLS & IMPLEMENTATION COST DETAILS	COST ALLOCATION (PHP)
Edge Telemetry Probe Hardware	Industrial fanless Quad-Core x86 Micro-PC (Dual GbE, 8GB RAM, DIN-rail mount) for local edge ICMP/eBPF probe exports	₱8,500.00
Controller Cloud / Server Allocation	Amortized annual server instance (Docker, Prometheus, Grafana, TimescaleDB, alerting broker) allocated per branch node	₱2,400.00
Engineering Manpower (Setup & Config)	Network automation & DevOps engineer labor (WAN router integration, BGP/tunnel failover testing in EVE-NG testbed, 16 hrs @ ₱350/hr)	₱5,600.00
On-Site Commissioning & QA	Field technician site visit, physical rack mounting, failover threshold validation, and SME handover	₱2,500.00
Administrative & Documentation	SOP user manual, compliance checklists, tamper-evident security labels, and warranty buffer	₱1,000.00
	TOTAL COST OF DEPLOYMENT PER BRANCH SITE:	**₱20,000.00**

B. ENTERPRISE TARGET PRICE DETERMINATION & LICENSING MODEL

We benchmark against proprietary SD-WAN vendors (Cisco Meraki, Fortinet, Silver Peak) which charge over ₱60,000–₱120,000 per branch annually. Our vendor-agnostic framework provides a highly competitive alternative:

OFFERING TIER	TARGET PRICE	UNIT COGS	GROSS PROFIT	GROSS MARGIN
Per-Branch Appliance + 1-Yr License	**₱35,000.00**	₱20,000.00	**₱15,000.00**	**42.9%**
Annual Enterprise Support Renewal (Year 2+)	**₱8,500.00/yr**	₱2,400.00/yr	**₱6,100.00/yr**	**71.8%**
Turnkey 20-Branch Enterprise Rollout	**₱650,000.00**	₱380,000.00	**₱270,000.00**	**41.5%**

C. MARKET SIZING CALCULATIONS (CAPSTONE PROJECT: ENTERPRISE WANS)

TAM • PHILIPPINE ENTERPRISE WANS

₱2.80 Billion

80,000 total branch sites across government agencies, banks, BPOs, and retail chains nationwide × **₱35,000** per branch deployment.

SAM • PUBLIC AGENCIES & MEDIUM BPOS

₱420.0 Million

12,000 distributed regional/district offices of national government agencies and regional mid-sized IT/BPO offices operating hybrid dual-ISP links × **₱35,000**.

SOM • INITIAL 3-YEAR AGENCY BEACHHEAD

₱10.5 Million

300 branch sites across 3–5 public agency partners (like our capstone target agency with 200+ district offices) × ₱35,000 = **₱10.5M**, plus recurring SLA renewals.

04 COMPARATIVE SYNTHESIS: B2C IOT VS. B2B ENTERPRISE INFRASTRUCTURE

DIMENSION	TECHNO PROJECT: HULOG (IOT VAULT)	CAPSTONE PROJECT: SELF-HEALING HYBRID WAN
Market Nature	**B2C / Consumer & Micro-Rental:** High volume, mass residential e-commerce buyers & landlords.	**B2B / Government Enterprise:** Low volume, high contract value, multi-office branch networks.
Production Cost (COGS)	**₱2,000.00** (₱1,700 materials + ₱180 labor + ₱120 overhead).	**₱20,000.00** per branch (₱8.5k probe + ₱5.6k engineering + ₱2.4k cloud/server + ₱3.5k field/QA).
Target Selling Price	**₱3,499.00** (42.8% gross margin) + ₱199/mo SaaS.	**₱35,000.00** per branch (42.9% gross margin) + ₱8.5k/yr SLA.
TAM Market Size	**₱38.5 Billion** (11.0M online buying households).	**₱2.80 Billion** (80,000 enterprise branch sites in PH).
SAM Target Segment	**₱5.25 Billion** (1.5M Mega Manila gated homes & dorms).	**₱420.0 Million** (12,000 government & BPO hybrid WAN branches).
3-Year SOM Goal	**₱35.0 Million** (10,000 units sold @ 0.65% SAM share).	**₱10.5 Million** (300 branch deployments across 3–5 agencies).

DIMENSION	TECHNO PROJECT: HULOG (IOT VAULT)	CAPSTONE PROJECT: SELF-HEALING HYBRID WAN
Adoption Bottleneck	Gate installation physical mounting & user habit shift.	Government procurement bidding cycles (RA 9184) & security approval.

Key Takeaway: Hulog relies on a *high-volume, consumer distribution model* where affordability (sub-₱3,500 retail) and rapid self-installation drive adoption. Conversely, our Capstone project operates on a *high-value enterprise systems model* where reliability, vendor-independence, and uptime preservation justify higher per-site integration pricing. Both maintain a healthy **~42% to 43% gross margin**, ensuring strong long-term commercial sustainability.

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